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Notifications 

4. Solving first order systems

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Numerically solving systems of first order ODEs



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and a vector containing the initial values of the S-I-R variables.

Make sure to match the order used in the derivative function.

After we run this script, the solution to our differential equation is stored in the output variables 'tSol' and 'xSol'.

The first output is a vector that contains the time values at which the numerical solution was computed.

The second output is a matrix, which contains the computed values of the variable 'x' for each time in 'tSol'. Each column of the matrix

Solving higher order systems with ODE45

In general, to solve a n -dimensional system using ODE45 in MATLAB, the following syntax is used:

$$\underbrace{[t, x]}_{\text{outputs}} = \text{ode45}(\underbrace{@(t, x) \text{odefun}}_{\text{inputs}}, \text{tspan}, x_0)$$

ODE45 requires that you provide the following three inputs:

- @(t,x) odefun:** **odefun** is a $n \times 1$ column vector $[f_1(t, x); f_2(t, x); \dots; f_n(t, x)]$ specifying the ODE to be solved. The notation **@(t,x)** is used to tell MATLAB that each f_i is a function of t and x .
- tspan:** A 1×2 row vector $[t_0, t_{\text{end}}]$ specifying the start and end times between which to solve the system of DEs.
- x0:** A $n \times 1$ column vector specifying the initial conditions $x(t_0)$.

In this course we will use the default settings of **ODE45** but additional inputs can be provided to control parameters such as the error tolerance and maximum step size. More details are can be found in the [MATLAB documentation](#).

After it has finished numerically solving the DEs, **ODE45** will return two outputs:

- t:** A column vector $[t_1, t_2, \dots, t_p]$ containing the times t_i at which the numerical solution has been computed.
- x:** A matrix containing n columns $[x_1, x_2, \dots, x_n]$, where x_i is a column vector containing the values of the solution x_i at the times in the vector

containing the values of the solution $\mathbf{x}$, at the times in the vector $\mathbf{t}$.

Ungraded exploration

Use MATLAB Online to solve the system from the video for different (positive) initial conditions. Try plotting your solutions!

4. Solving first order systems

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